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To cite this article: Amit Kumar, Gaurav Bharadwaj, Aayush Gupta & Kamal Sharma (2024) Correlation study of thermal charging, discharging, and efficiency of graphene oxide-reinforced paraffin wax for thermal energy storage system, Chemical Engineering Communications, 211:9, 1336-1342, DOI: [10.1080/00986445.2024.2341275](https://doi.org/10.1080/00986445.2024.2341275)

To link to this article: <https://doi.org/10.1080/00986445.2024.2341275>



Published online: 12 Apr 2024.



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Correlation study of thermal charging, discharging, and efficiency of graphene oxide-reinforced paraffin wax for thermal energy storage system

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ABSTRACT

Nanocomposites, produced through mechanical blending, serve to enhance thermal performance of heat exchangers, solar collectors, and photovoltaic-thermal systems. This work employs an experimental methodology to assess the thermal charging (TC) and thermal discharging (TDC) of a composite generated by mixing graphene oxide (GO) and paraffin wax. TC and TDC of the GO-reinforced paraffin wax-based phase change material (PCM) have been examined across different flow rates (0.375, 0.75, and 1.5 L/min) and concentrations of GO (ranging from 0.25 to 1.00 vol%). The results exhibit a direct correlation between TC, TDC and efficiency, and reaction parameters (flow rate and volume concentration of GO). The experimental and predicted values of TCs, TDCs, and efficiencies fell $\pm 15\%$ (for TC and TDC), and $\pm 9\%$ for thermal efficiency, demonstrating a satisfactory level of agreement. This holds a practical application and valuable predictive tool for estimating TC, TDC, and thermal efficiency on known flow rates and concentrations of GO in wax-based PCMs.

KEYWORDS

Correlation study; graphene oxide; paraffin wax; thermal charging; thermal discharging

1. Introduction

In a world increasingly reliant on intermittent renewables, thermal energy storage (TES) systems are vital for effective management of energy resources, enhancing reliability, and decreasing dependence on fossil fuels (Guangul and Chala 2019). TES comprises a variety of methods, including temperatures adjustment for sensible heat storage, phase shifts in latent heat storage, reversible chemical reactions in thermochemical storage, and heat pump systems for effective heat extraction (Gonzalez-Nino et al. 2018; Pisani 2020; Seddegh et al. 2018). The application of PCMs (e.g., paraffin wax and fatty acids, etc.) in TES systems enhances the effectiveness of energy storage and release, resulting in increased energy efficiency and reduced reliance on conventional sources. This is especially advantageous for tasks like regulating building climate and incorporating renewable energy (Javadi et al. 2020; Jiang et al. 2019; B. Xu et al. 2015). The integration of PCMs with GO forms a composite with enhanced thermal properties and great potential for various

industries contributing to a more sustainable and energy-efficient future (Wang et al. 2014; Zeng et al. 2012). Jiang et al. (2023), Liu et al. (2016, 2020), and Liu and Yang (2017) investigated the hydrate salt-based PCMs for thermal applications which were modified by using TiO_2 and GO to enhance its chemical and thermal stability in TES systems. While Kalidasan et al. (2022, 2023) reported the effect of graphene on chemical, thermal, and optical properties of two different types of PCMs (i.e., organic paraffin and eutectic inorganic salt hydrate) for TESS. Conducting a meticulous correlation study of TES systems is crucial for validating models and ensuring efficient energy capture and release. Brousseau and Lacroix (1998) introduced a computational model predicting the thermal behavior of multi-layer latent heat storage units, unveiling intricate isothermal distributions, and highlighting crucial factors impacting heat load and energy storage. Motahar et al. (2017) discussed the solidification process of n-octadecane with different TiO_2 concentrations, underscoring their viability as TES

mediums for enhancing building comfort. Guo and Zhang (2008) proposed a mathematical approach to measure the effectiveness of Al foils to enhance heat transfer and discharging processes in high-temperature latent heat thermal energy storage systems (LHTES). A novel correlation has also been reported for quick and accurate simulation of discharging time in plate-based LHTES systems, addressing a key challenge in their commercial deployment and design complexity (Diarce et al. 2018). Also, a novel dimensionless total melting time concept for LHTES units has been proposed, with a correlation formula that accounts for various parameters and natural convection effects, demonstrating lower-order nonlinearity compared to existing methods (Y. Xu et al. 2019). Mekrisuh et al. (2020) examined flow and geometric factors influencing total melting time in a concentric double-pipe TES system using PCMs, developing a comprehensive correlation with $\pm 15\%$ accuracy. Gautam and Saini (2021) established correlations for Nusselt number and friction factor in a spherical-shaped packing element-based packed bed storage system for low-temperature solar thermal energy applications, offering predictive accuracy of around 6.7–7.5%.

Till date, several articles have addressed TC, TDC, and overall efficiency in TES systems, often using different materials to improve the properties of paraffin wax (Jiang et al. 2023; Kalidasan et al. 2022, 2023; Liu et al. 2016, 2020). But the mathematical correlation between thermal performance (TC, TDC, and overall efficiency) and process parameters of wax-based PCMs have not been reported yet. Therefore, the manuscript describes the step wise correlation study as an effect of flow rate (heat transfer fluid [HTF]) and GO concentration in paraffin wax on the TC, TDC, and thermal efficiency.

2. Experimental section

2.1. Sample preparation

Paraffin wax and GO were procured from Ecosense Sustainable Solutions Pvt. Ltd., India and Platonic Nanotech Pvt. Ltd., India (respectively). The thermophysical properties of the

Table 1. Thermophysical properties of paraffin wax as PCM

Properties	Magnitude
Density (kg/m^3)	900
Latent heat of fusion (kJ/kg)	210
Thermal conductivity (W/m.K)	0.2
Melting temperature ($^{\circ}\text{C}$)	56–60
Specific heat (J/kg.K)	2.1

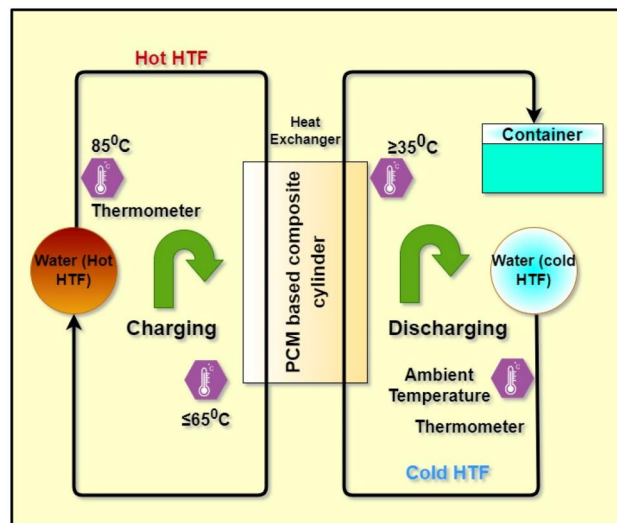


Figure 1. Flow diagram of experimental setup and data acquisition.

paraffin wax are given in Table 1. Paraffin wax-based nanocomposites were prepared employing various concentrations of GO (0.25, 0.50, 0.75, and 1.00 vol%) *via* mechanical blending (magnetic stirring) at a temperature of 65°C for 6 h. Then, the molten GO-based wax was filled in storage cylinder, mentioned in subsequent sections.

2.2. Experimental setup and calculations

The thermal study was conducted with an experimental setup as depicted in Figure 1. The setup included a storage cylinder (as heat exchanger) filled with prepared PCM-based composites, facilitating two distinct heat cycles: TC and TDC. In the charging (TC) cycle, water served as the HTF, heated to 85°C through electric heaters, and circulated through heat exchanger to transfer heat from HTF to composite samples. The successful melting of paraffin wax, at 65°C , confirmed the completion of charging cycle. The discharging (TDC) cycle involved cold HTF to extract heat energy from the heat exchanger. The cycle was concluded after achieving 35°C

Table 2. Charging and discharging time of GO-based paraffin wax nanocomposite at different flow rates

GO concentration (vol.%)	Overall charging time (min)			Overall discharging time (min)			Efficiency (%)		
	0.375 L/min	0.75 L/min	1.5 L/min	0.375 L/min	0.75 L/min	1.5 L/min	0.375 L/min	0.75 L/min	1.5 L/min
0.0	160	140	120	250	210	190	31.87	28.69	21.52
0.25	140	130	110	200	220	180	35.20	33.69	28.50
0.50	130	110	100	180	200	170	38.80	36.98	31.66
0.75	90	80	60	150	140	130	46.30	43.79	41.70
1.00	110	100	80	160	180	140	40.40	38.16	34.02

temperature of heat exchanger. Assumptions for analysis included incompressible, steady, and laminar fluid flow, negligible heat losses, constant thermal resistance of PCM cylinder wall, and temperature-independent properties of PCM and HTF. All the temperature sensors, shown in Figure 1 provided the data with <3.0% of error and the time measurements were recorded with the help of stopwatch. All the data are given in Table 2, recorded under different test conditions (GO concentration and flow rate of HTF).

The overall efficiency of PCM-based TES system has been determined by the ratio of output energy (E_{out}), transferred to cold HTF (from molten PCM), and input energy (E_{in}), transferred to PCM (from hot HTF). These energies and overall efficiency of TESS (without considering energy storage time) were computed with the help of following Equations (1)–(3), respectively.

$$\text{Charging; } E_{in} = \dot{m} C_{HTF} \int_0^t [T_{in,HTF} - T_{out,PCM}] dt \quad (i)$$

$$\text{Discharging; } E_{out} = \dot{m} C_{HTF} \int_0^t [T_{out,PCM} - T_{out,WF}] dt \quad (ii)$$

$$\text{Overall efficiency; } \eta = \frac{E_{out}}{E_{in}} \quad (iii)$$

where C_{HTF} is specific heat (4.18 kJ/kg.K) and $\dot{m}$ is mass flow rate (kg/s) of HTF. The overall charging, discharging and overall efficiency are given in Table 2.

3. Results and discussion

Correlation of TC, TDC, and efficiency with GO concentration (ϕ) and HTF flow rate (m) has been discussed in subsequent section.

3.1. Correlation study – charging time

A mathematical correlation has been developed using the experimental data for TC at different GO concentrations ($\phi = 0.25$ – 1.00 vol.%) and flow rate ($m = 0.375$, 0.75 , and 1.5 L/min). Figure 2(a) represents the ‘TC’ vs. ‘ m ’ in which regression analysis predicted the following linear equation to be followed,

$$t_c = -26.19m + B0 \quad (iv)$$

where $B0$ Y-intercept. Further, values of ‘ $B0$ ’ have been plotted against ‘ ϕ ’ as shown in Figure 2(b) and regression analysis was carried out to fit polynomial function as given below,

$$t_c + 26.19m = 133.33\phi^2 - 217.33\phi + 199.58 \quad (v)$$

From Equation (ii), the following correlation for TC (t_c) can be established as a function of ‘ ϕ ’ and ‘ m ’,

$$t_c = -133.33\phi^2 - 217.33\phi - 26.19m + 199.58 \quad (vi)$$

Equation (iii) is valid for the undertaken boundary condition of GO concentration, i.e., 0.25 – 1.00 vol% and flow rate, i.e., 0.375 – 1.5 L/min, only. Figure 2(c) shows the comparison of t_c (predicted) and t_c (experimental). It can be seen from the figure that the predicated reading and experimental values lie between the $\pm 15\%$ for GO concentration range from 0.25 to 1.00 vol% which is in the good acceptable limit.

3.2. Correlation study – discharging time

A mathematical correlation has been developed using the experimental data for TDC at different ($\phi = 0.25$ – 1.00 vol.%) and ($m = 0.375$, 0.75 , and 1.5 L/min). Figure 3(a) represents the ‘TDC’ vs. ‘ m ’ in which regression analysis predicted the following quadratic equation to be followed,

$$t_d = 17.778m^2 - 73.333m + C0 \quad (vii)$$

where $C0$ is Y-intercept. Further, values of ‘ $C0$ ’ have been plotted against ‘ ϕ ’ as shown in

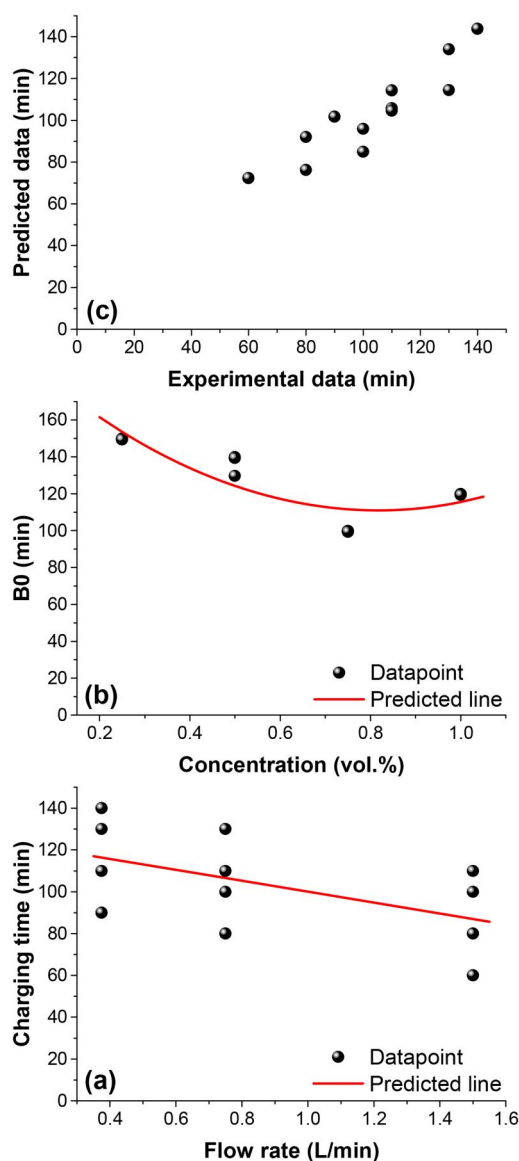


Figure 2. (a) Variation of charging time with respect to flow rate, (b) 'B0 vs. GO concentration', and (c) comparison of charging time (predicted and experimental).

Figure 3(b) and regression analysis was carried out to fit polynomial law as given below,

A regression analysis was carried out to fit polynomial function as given below,

$$t_d - 17.779m^2 + 73.333m = 106.67\phi^2 - 192\phi + 265 \quad (\text{viii})$$

From Equation (v), the following correlation for t_d can be established as a function of ' ϕ ' and ' m ',

$$t_d = 17.779m^2 - 73.333m + 106.67\phi^2 - 192\phi + 265 \quad (\text{ix})$$

Equation (vi) is valid for the undertaken boundary condition of GO concentration, i.e.,

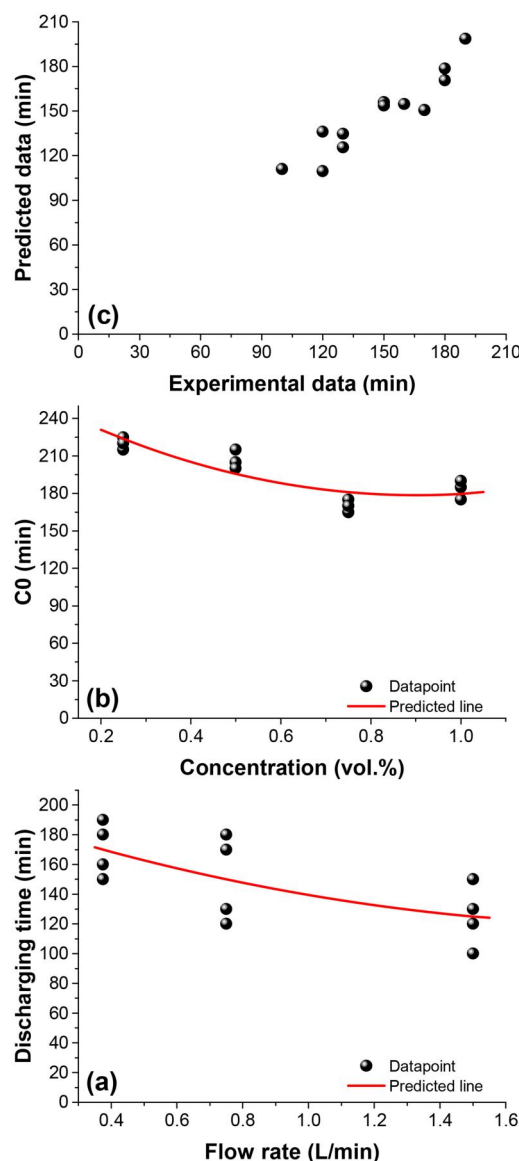


Figure 3. (a) Variation of discharging time with respect to flow rate, (b) 'C0 vs. GO concentration', and (c) comparison of discharging time (predicted and experimental).

0.25–1.00 vol% and flow rate, i.e., 0.375–1.5 L/min, only. Figure 3(c) shows the comparison of t_d (predicted) and t_d (experimental). It can be seen from the figure that the predicated reading and experimental reading lie between the $\pm 15\%$ for vol. concentration range from 0.25 to 1% which is in the good acceptable limit.

3.3. Correlation study – thermal efficiency

A mathematical correlation has been developed using the experimental data for efficiency at different ($\phi = 0.25$ –1.00 vol%) and ($m = 0.375$, 0.75, and 1.5 L/min). Figure 4(a) represents the

'efficiency' vs. ' ϕ ' in which regression analysis predicted the following quadratic equation to be followed,

$$\eta = -39.013\phi^2 + 58.089\phi + A0 \quad (x)$$

where $A0$ is the Y-Intercept. Further, values of ' $A0$ ' has been plotted against ' ϕ ' as shown in Figure 4(b) and regression analysis was carried out to fit polynomial law as given below,

$$\eta + 39.013\phi^2 - 58.089\phi = -5.5248m + 24.249 \quad (xi)$$

From Equation (viii), the following correlation for efficiency can be established as a function

of ' ϕ ' and ' m ',

$$\eta = -5.5248m - 39.013\phi^2 + 58.089\phi + 24.249 \quad (xii)$$

Equation (ix) is valid for the undertaken boundary condition of GO concentration, i.e., 0.25–1.00 vol% and flow rate, i.e., 0.375–1.5 L/min, only. Figure 4(c) shows the comparison of efficiency (predicted) and efficiency (experimental). It can be seen from the figure that the predicted reading and experimental reading lie between the $\pm 9\%$ for vol. concentration range from 0.25% to 1.00 vol% which is in the good acceptable limit.

4. Conclusion

The correlations between thermal performance (charging time, discharging time, and thermal efficiency) and process parameters (concentrations of GO in composites; 0.25–1.00 vol%, and mass flow rates; 0.375–1.5 L/min) have successfully been established.

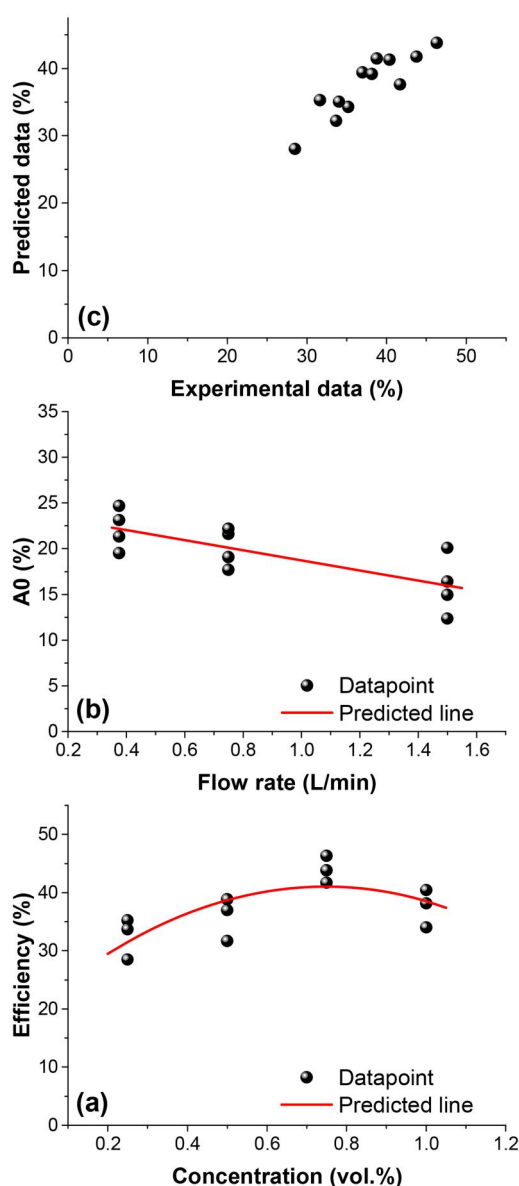


Figure 4. (a) Variation of efficiency with respect to GO concentration, (b) ' $A0$ vs. flow rate', and (c) comparison of efficiency (predicted and experimental).

- For charging of TES system, a regression analysis was conducted to derive two correlations. (i) linear relationship between ' t_c and flow rate', and (ii) ' t_c ' can be expressed as a polynomial function of flow rate and GO concentration, both. Following the similar approach, the t_d can be expressed with the help of polynomial function (order = 2) in terms of flow rate and GO concentrations.
- For thermal efficiency, a polynomial equation relating it to GO concentration was established followed by a regression analysis incorporating a linear term for flow rate alongside GO concentration.
- Experimental and predicted values of TCs, TDCs, and efficiencies fell within $\pm 15\%$ for charging and discharging times, and within $\pm 9\%$ for thermal efficiency, demonstrating a satisfactory level of agreement.
- These correlations hold significance for practical applications within the specified range of concentrations (0.25–1.00 vol%) of GO in paraffin wax-based composite. They offer a valuable predictive tool for estimating charging time, discharging time, and thermal efficiency based

on known mass flow rates and volume concentrations.

- The study's meticulous step-wise methodology ensured robustness in the development of these correlations, providing a reliable framework for further investigations and real-world applications in systems involving similar parameters.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Department of Science and Technology, Ministry of Science and Technology, India.

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